What Makes It Balance?

Name: _____ Period: _____

Solve each equation, then compare with the slider.

1 $2x + 3 = x + 7$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

2 $3x + 2 = x - 4$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

3 $x + 5 = 4x + 1$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

4 $2x + 3 = 2x + 8$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

5 $4x + 2 = 4x - 6$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

6 $3x - 5 = 3x - 5$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

What Makes It Balance?

Name: _____ Period: _____

Solve each equation, then compare with the slider.

7 $5 + x + x = 2x + 5$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

8 $5x + 2 = 5x + 3$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

9 $3x + 4 = 4$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

10 $2x + 3 + x = 3x + 3$

Balanced: Sometimes / Always / Never
circle one

Balanced when: x = _____ / x is any number / not possible
circle one

Solve it:

Result: _____ = _____

11 $3x - 6 + x = 4x + 1$

My prediction: Sometimes / Always / Never
circle one
because _____

Solve it:

Result: _____ = _____

12 $5x - 2 = 2x + 10$

My prediction: Sometimes / Always / Never
circle one
because _____

Solve it:

Result: _____ = _____